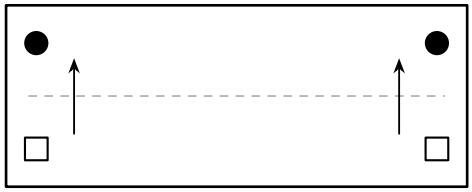
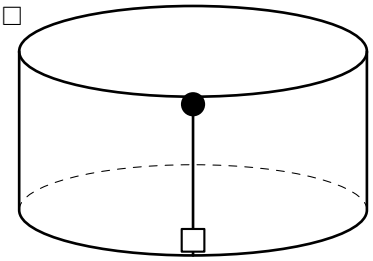
Keep the bands on the table. An adult makes every cut.

input process output



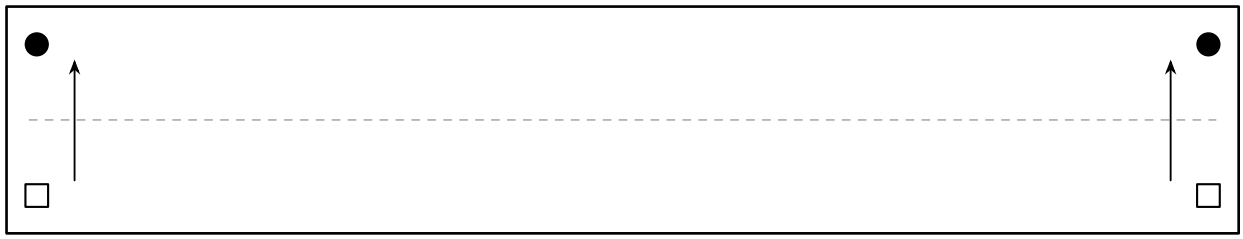
Join • to • and □ to □.



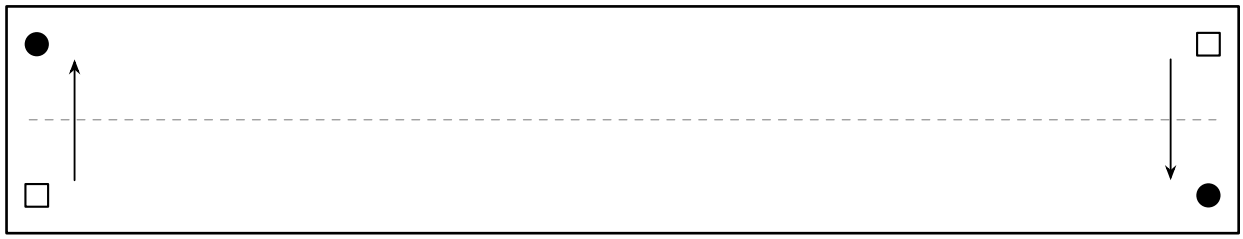
An edge trip stays on an edge, follows its seam marks, and stops on its first return to the start dot.

**Problem 1:** Trace all the edges of A and B. How many separate closed edges does each band have?

A

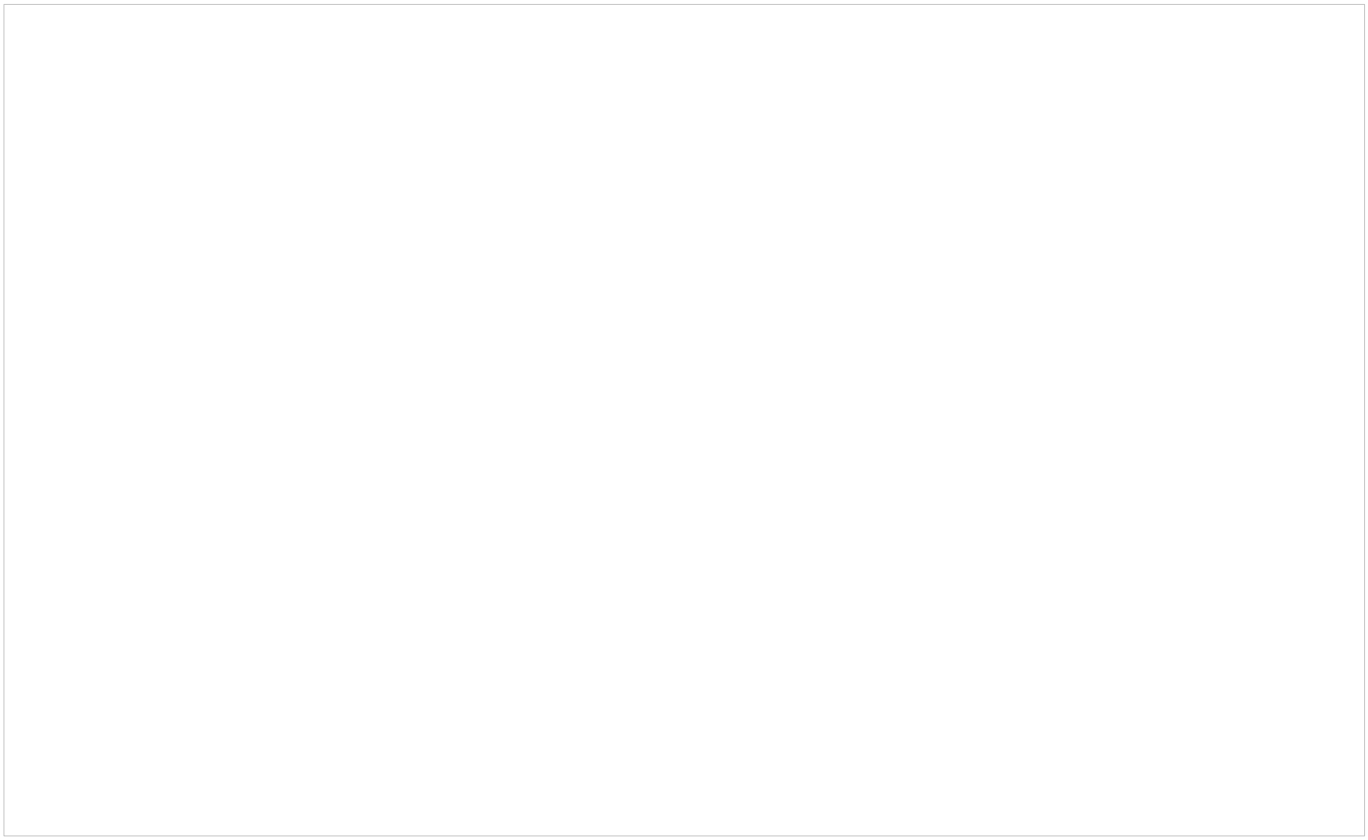


B

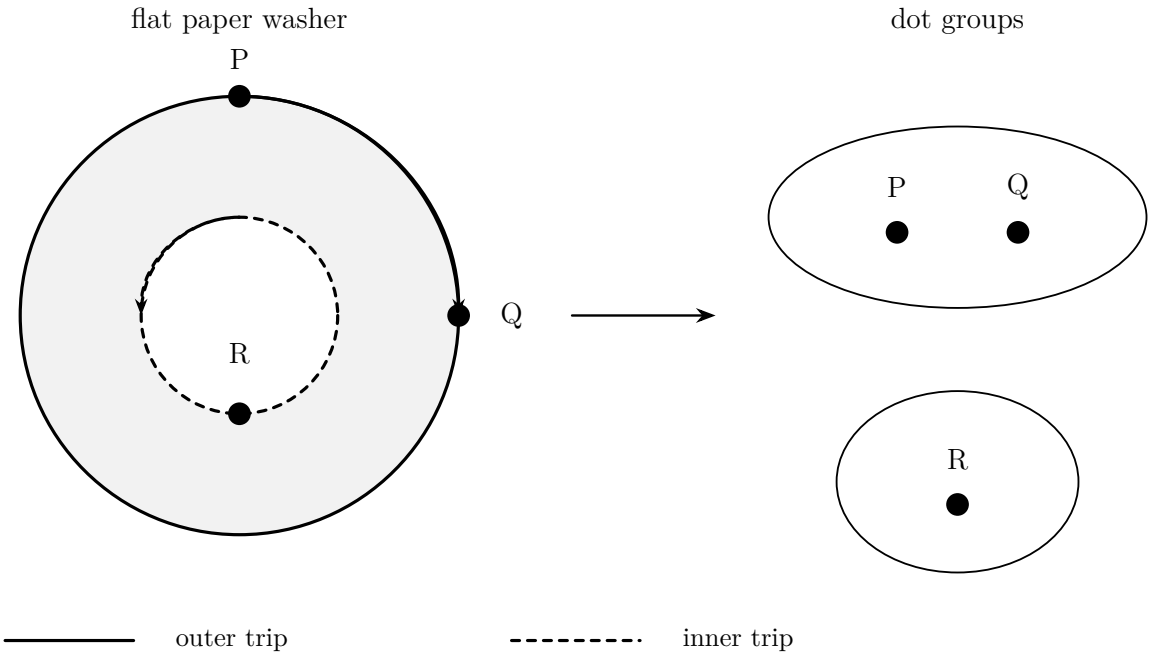


A: _____ B: _____

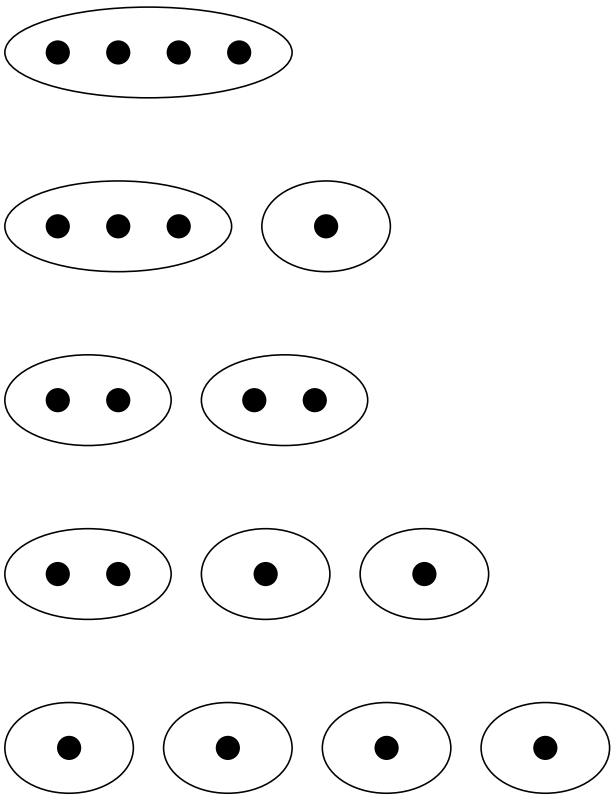
**Problem 2:** On A and B, find two edge dots that one edge trip can visit together, and two that it cannot. Which requests are impossible?



Dots on one edge go in one ring.




**Problem 3:** Place four edge dots to match each picture on A and on B. Write yes or no in each blank. Each ring means a different closed edge; other edges may have no dots.



A: _____ B: _____

A: _____ B: _____

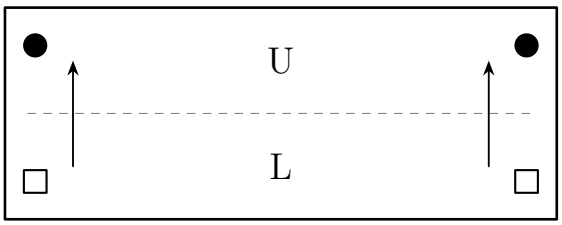
A: _____ B: _____

A: _____ B: _____

A: _____ B: _____

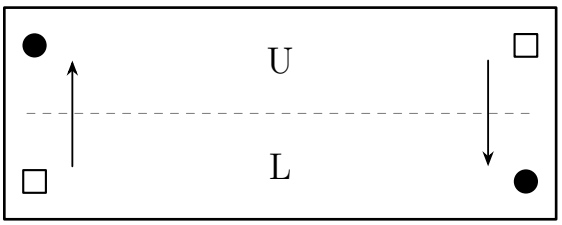
**Problem 4:** Predict how many connected pieces a middle-line cut will make in A and B. Ask an adult to make both cuts and record what happened.

A




I predict: _____

B



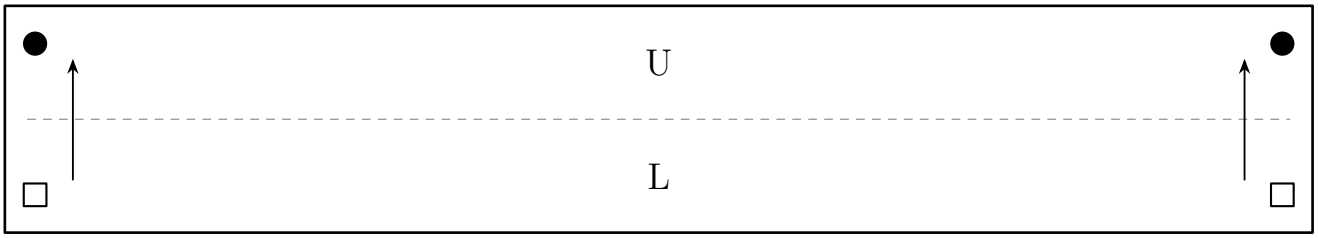

I predict: _____

after cutting A

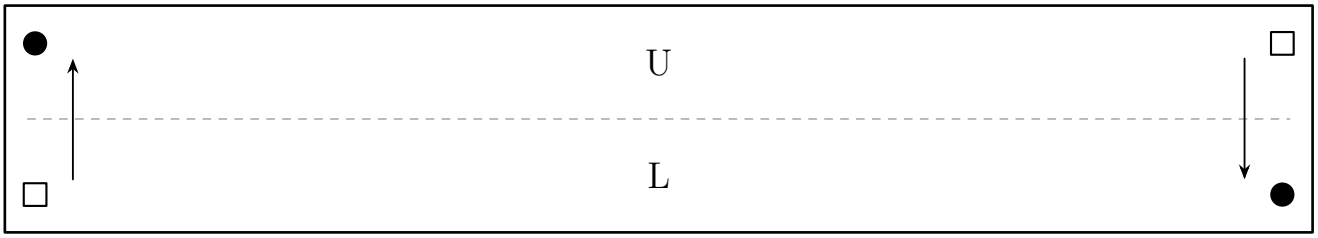
after cutting B

**Problem 5:** Use the U and L marks to explain why the cuts made that many pieces. On each cut result, can two edge dots lie on the same piece but on different closed edges?

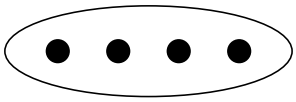
A



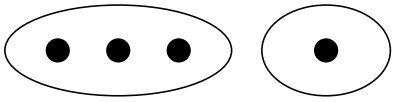

B



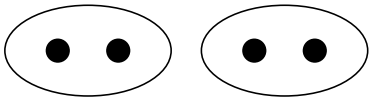
**Problem 6:** Use all the cut pieces from A together, then all those from B. Which four-dot pictures from Problem 3 can you now make? Write yes or no in each blank.



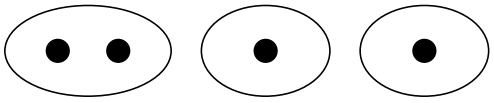
A: _____ B: _____



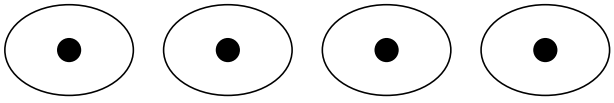
A: _____ B: _____



A: _____ B: _____



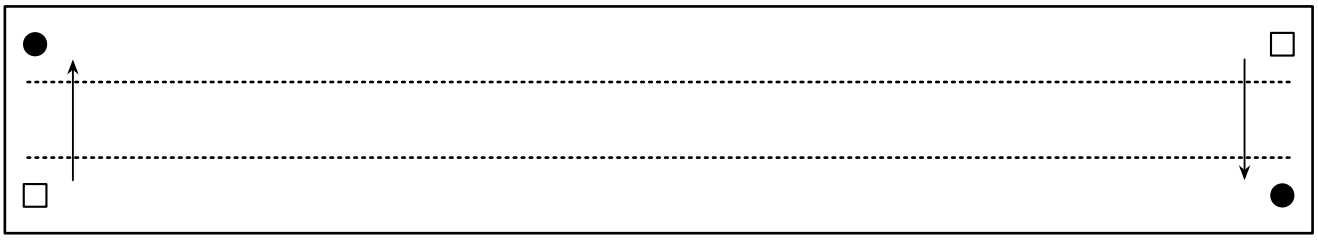
A: _____ B: _____

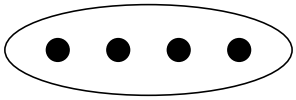


A: _____ B: _____

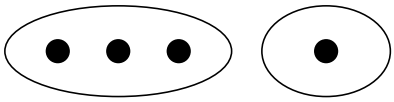
**Problem 7:** An adult cuts a new B one third of the way across, going round until the cut closes. Which four-dot pictures can you make now? Write yes or no in each blank.

B

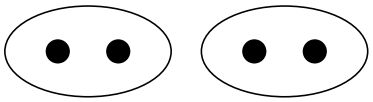




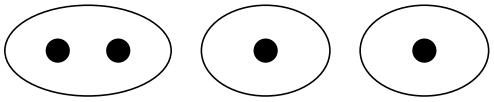
new B: _____



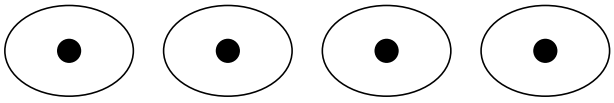
new B: _____



new B: _____



new B: _____



new B: _____